

Beyond Detection Parity: Localization Parity in Pedestrian Detection

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Abstract. Pedestrian detection is a safety-critical component of autonomous driving and intelligent surveillance, where equitable performance across demographic groups is essential for trustworthy deployment. Fairness research in this domain has evaluated performance almost exclusively through detection recall, that is, whether a person is found at all, leaving the precision of the predicted bounding box unexamined. Because existing audits reduce localization to a binary hit-or-miss threshold, systematic disparities in how tightly detectors localize different groups remain invisible, even though loose boxes directly compromise downstream distance and trajectory estimates. This paper introduces *Localization Parity*, a fairness criterion that requires equal expected Intersection over Union across demographic groups conditional on detection. We present the first confounder-controlled audit spanning four driving datasets (BDD100K, CityPersons, EuroCity-Day, and EuroCity-Night), three detector families (YOLOX, RetinaNet, and Cascade R-CNN), and two demographic axes (gender and age), together with a demographic-weighted bounding box regression intervention. Female and child pedestrians receive significantly looser boxes in all twelve dataset and detector combinations, with mean gaps of 1.40 and 3.27 percentage points. These gaps survive scale stratification in 35 of 36 and 24 of 30 cells, respectively. The proposed intervention closes the training-set gap by 46.8%, yet fails to generalize to validation data. Localization Parity therefore exposes a previously unmeasured and safety-relevant axis of algorithmic unfairness, and the resulting fairness generalization gap poses a concrete open problem for equitable perception systems.

Keywords: Localization Parity · Algorithmic Fairness · Pedestrian Detection · Bounding Box Regression · Demographic Bias · Object Detection

1 Introduction

Pedestrian detection is central to modern computer vision pipelines for self-driving cars, surveillance, and traffic infrastructure. The cost of error is not

shared evenly across demographic groups. Prior work has shown that *detection recall* is unequal: Wilson et al. [14] report lower average precision for darker-skinned pedestrians, while Brandao [2] and Li et al. [9] find substantially higher miss rates for children than for adults across many detectors.

What these audits leave unmeasured is *localization quality*. Once a detector draws a box, the tightness of that box shapes every estimate that follows: position, distance to the ego vehicle, predicted trajectory, and braking urgency. Two detections can both pass the conventional $\text{IoU} \geq 0.5$ hit threshold and still differ sharply in geometry. A box at IoU 0.51 and a box at IoU 0.91 look identical under recall-based fairness metrics, yet they feed very different signals into downstream modules. If one demographic group systematically receives looser boxes, that group systematically receives worse downstream estimates, and no current fairness benchmark would notice.

We formalize this overlooked criterion as **Localization Parity (LP)**: equality of expected Intersection over Union (IoU) across demographic groups, conditional on successful detection and after controlling for confounders such as object scale. We then run the first multi-dataset, multi-detector LP audit and test a training-time mitigation.

Contributions.

- A formal definition of Localization Parity as a group fairness criterion on box regression quality, orthogonal to detection parity and linked to existing fairness notions for classification and regression.
- The first confounder-controlled LP audit across four autonomous driving datasets, three detector families spanning single-stage and two-stage designs, and two demographic axes: gender (female vs. male) and age (child vs. adult).
- Empirical evidence that female and child pedestrians receive consistently lower IoU boxes (mean gaps -1.40 pp and -3.27 pp), surviving ICON² scale stratification [13].
- Demographic-Weighted Bounding Box Regression (DWBBR), which closes the training gap by 46.8% yet fails to transfer cleanly to held-out data, documenting a *fairness generalization gap* as an open problem.

2 Related Work

Fairness research in machine learning has developed a mature vocabulary of group criteria, including demographic parity and equalized odds [7], individual fairness, and calibration. In computer vision, these ideas are usually operationalized through equal detection rates or equal average precision. Wilson et al. [14] established predictive inequity across Fitzpatrick skin-tone groups on BDD100K using AP, while Schumann et al. [12] caution that binary third-party gender labels can themselves encode annotation bias. Fair regression [1] constrains disparities in continuous prediction error; Localization Parity applies the same continuous-loss perspective to the IoU of matched detections. What this

literature still lacks is a criterion that isolates box geometry after a detection has already been accepted.

Pedestrian detection fairness has likewise focused on hit-or-miss outcomes. BDD100K [15], CityPersons [16], and EuroCity Persons [3] remain the dominant driving benchmarks. Brandao [2] and Li et al. [9] report elevated child miss rates across many detectors, and Li et al. further observe higher female miss rates under low light. Khoshkdahan et al. [8] study pose and occlusion as structural confounders of detection accuracy. Detector design improvements such as cascade refinement [4] and focal reweighting [10] improve overall box precision, yet neither was designed to equalize IoU across demographic groups. Confounder-controlled auditing via ICON² [13] and fair training via Group DRO [11] provide useful tools, but they have been applied mainly to classification or AP rather than to continuous localization quality.

Table 1 situates prior work against the present study. Existing audits measure detection parity (recall or AP), typically on one or two datasets, and either omit localization entirely or reduce it to a binary IoU threshold. No prior study defines Localization Parity, audits continuous IoU gaps across a multi-dataset and multi-detector matrix under scale stratification, and tests a regression-level mitigation with an out-of-sample fairness check. That combination is the research gap this paper addresses.

Table 1. Comparison with representative prior work. DP: detection parity (recall/AP). LP: continuous localization parity (mean IoU). CC: confounder control. Mit.: training-time mitigation of localization disparity.

Work	Metric	Datasets	Detectors	Axes	CC	Mit.
Wilson et al. [14]	AP	1	several	skin tone	partial	–
Brandao [2]	miss rate	1	24	age, gender	–	–
Li et al. [9]	miss rate	several	8	age, gender	–	–
ICON ² [13]	AP	1	several	skin tone	yes	–
Khoshkdahan et al. [8]	detection	1	several	pose/occ.	yes	–
This work	LP (IoU)	4	3	gender, age	yes	DWBBR

3 Proposed Solution

3.1 Localization Parity

Let $\mathcal{D} = \{(b_i, \hat{g}_i, g_i)\}$ be the set of successfully detected instances, where b_i is the predicted box, $\hat{g}_i$ the ground-truth box, and g_i the demographic group of instance i . Matching follows the COCO convention: a prediction counts as a detection only if $\text{IoU}(b_i, \hat{g}_i) \geq 0.5$. Localization Parity requires

$$\mathbb{E}[\text{IoU}(b_i, \hat{g}_i) \mid g_i = g_a] = \mathbb{E}[\text{IoU}(b_i, \hat{g}_i) \mid g_i = g_b] \quad (1)$$

for every pair of groups $a \neq b$, after controlling for instance scale. We report the signed gap $\Delta\text{IoU} = \mathbb{E}[\text{IoU} \mid g_{\text{priv}}] - \mathbb{E}[\text{IoU} \mid g_{\text{dis}}]$, where g_{priv} denotes the privileged group (male, adult) and g_{dis} the disadvantaged group (female, child). A negative gap means the disadvantaged group receives worse boxes. LP is deliberately conditional on detection: it equalizes box quality among people who were found, and is therefore orthogonal to detection parity. Relative to classification fairness criteria, LP replaces a binary outcome with a continuous quality score; relative to fair regression, it uses the loss $\ell = 1 - \text{IoU}$ on matched boxes; relative to AP-based audits, it isolates the localization component that AP mixes with confidence ranking.

Gaps of 1–3 pp IoU are practically consequential. Under a pinhole projection, relative depth error scales with relative box-height error. For a pedestrian at 30m whose true box height is 80 px, a 2 pp mean IoU deficit (roughly 2–4 px of systematic height or vertical-center error) translates into about 0.7–1.5 m of depth bias. That is enough to alter time-to-collision estimates at urban speeds and to shift predicted crossing trajectories, so we treat LP gaps in this range as safety-relevant.

3.2 Demographic-Weighted Bounding Box Regression

DWBRR applies the Group DRO reweighting principle [11] to the CIoU regression branch. Standard YOLO training minimizes $L_{\text{class}} + \lambda L_{\text{reg}}$. DWBRR replaces the regression term with a group-weighted sum

$$L_{\text{DWBRR}} = L_{\text{class}} + \lambda \sum_i w_{g_i} L_{\text{CIoU}}(b_i, \hat{g}_i), \quad (2)$$

where $w_g \propto 1/\mathbb{E}_{\text{train}}[\text{IoU} \mid g]$ and weights are renormalized each epoch so that $\sum_g w_g = 1$. No architecture change is required: we override the Ultralytics training callback to recompute weights from per-group training CIoU and scale the regression loss before backpropagation.

3.3 Experimental Design

Figure 1 summarizes the overall study. Phase 1 audits Localization Parity; Phases 2 and 3 test whether DWBRR closes the gap and whether the improvement transfers out of sample.

We evaluate on four pedestrian benchmarks chosen for geographic, sensor, and lighting diversity: BDD100K [15] (large-scale US dashcam data with about 100,000 annotated pedestrian boxes and binary male/female labels), CityPersons [16] (about 19,000 boxes over 2,975 Cityscapes images with careful occlusion annotation), and EuroCity-Day together with EuroCity-Night [3] (subsets of EuroCity Persons, covering about 238,200 boxes from 31 European cities). Demographic labels for CityPersons and EuroCity use augmented annotations where available; detections with missing or ambiguous labels are dropped.

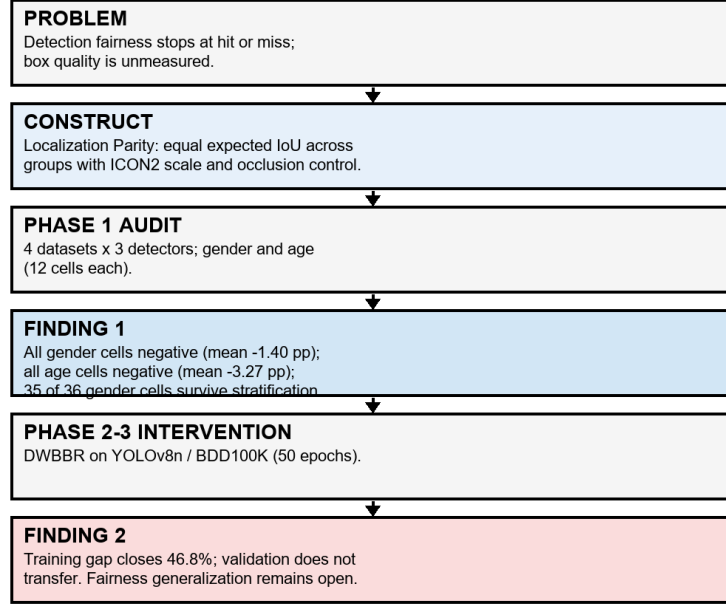


Fig. 1. Conceptual framework. Phase 1 audits Localization Parity across four datasets and three detectors under ICON² confounder control. Phase 2 and Phase 3 test whether DWBBR closes the gap and reveal a fairness generalization barrier.

The audit covers three detector families with public pretrained weights and no fine-tuning: YOLOX [6] (single-stage, anchor-free), RetinaNet [10] (single-stage, anchor-based), and Cascade R-CNN [4] (two-stage, with cascaded IoU refinement). The mitigation experiments use YOLOv8n as a lightweight trainable backbone. The two protected attributes are gender (female vs. male) and age (child vs. adult). Race and skin tone are excluded because an earlier pilot lacked sufficient labeled sample size and used categories too coarse for reliable inference.

Predictions are matched to ground truth by greedy, confidence-sorted IoU assignment; matches with $\text{IoU} \geq 0.5$ enter the LP analysis. The primary confounder is object scale (box height), because female and child pedestrians tend to occupy smaller image regions. Following ICON², we bin ground-truth instances within each dataset into short (≤ 33 rd height percentile), medium (34th–66th), and tall (≥ 67 th) strata, then recompute per-group mean IoU inside each bin. Within each cell and stratum we compare group IoU distributions with the two-sided Mann–Whitney U test at $p < 0.05$.

Both DWBBR and an identically configured baseline (unweighted YOLOv8n) are trained on a 1,693-image BDD100K subset with reliable gender labels (64.8%

male and 35.2% female), for 50 epochs, batch size 8, image size 416×416 , and $\text{lr}_0 = 0.01$ with cosine decay on a single NVIDIA RTX 3050. Validation uses a disjoint 423-image subset. We replicate with seeds $\{42, 123, 7\}$ and select the checkpoint with the best overall validation CIoU unless otherwise noted.

4 Experimental Results

4.1 Localization Parity Audit

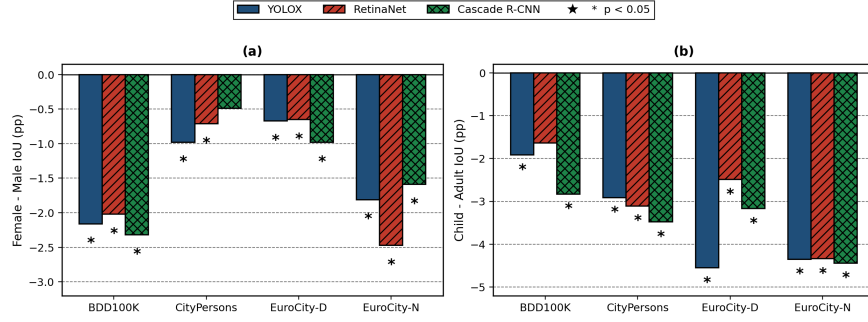
Female pedestrians receive lower mean IoU than male pedestrians in all twelve dataset and detector cells (mean gap -1.40 pp; eleven of twelve significant). The lone non-significant cell is CityPersons with Cascade R-CNN (-0.49 pp, $p = 0.11$). BDD100K shows the most consistent gaps (-2.16 , -2.02 , and -2.32 pp), and EuroCity-Night is comparably severe. Children receive lower mean IoU than adults in all twelve cells as well (mean -3.27 pp; eleven of twelve significant), roughly double the gender gaps, with EuroCity-Night again the harshest setting. Tables 2 and 3 and Figure 2 report every cell. Under ICON² scale stratification the gender gap survives in 35 of 36 cells (mean -1.43 pp) and the age gap in 24 of 30 within-scale cells, so the disparities are not reducible to women and children occupying smaller image regions.

Table 2. Phase 1 LP audit, gender axis. Gap = female minus male mean IoU (pp). †: $p < 0.05$, Mann–Whitney U .

Dataset	Detector	Gap (pp)
BDD100K	YOLOX	$-2.16^\dagger$
BDD100K	RetinaNet	$-2.02^\dagger$
BDD100K	Cascade R-CNN	$-2.32^\dagger$
CityPersons	YOLOX	$-0.98^\dagger$
CityPersons	RetinaNet	$-0.71^\dagger$
CityPersons	Cascade R-CNN	-0.49
EuroCity-Day	YOLOX	$-0.67^\dagger$
EuroCity-Day	RetinaNet	$-0.65^\dagger$
EuroCity-Day	Cascade R-CNN	$-0.98^\dagger$
EuroCity-Night	YOLOX	$-1.81^\dagger$
EuroCity-Night	RetinaNet	$-2.47^\dagger$
EuroCity-Night	Cascade R-CNN	$-1.59^\dagger$
Mean		-1.40

Table 3. Phase 1 LP audit, age axis. Gap = child minus adult mean IoU (pp). †: $p < 0.05$, Mann–Whitney U .

Dataset	Detector	Gap (pp)
BDD100K	YOLOX	−1.92 [†]
BDD100K	RetinaNet	−1.64
BDD100K	Cascade R-CNN	−2.83 [†]
CityPersons	YOLOX	−2.91 [†]
CityPersons	RetinaNet	−3.11 [†]
CityPersons	Cascade R-CNN	−3.48 [†]
EuroCity-Day	YOLOX	−4.55 [†]
EuroCity-Day	RetinaNet	−2.49 [†]
EuroCity-Day	Cascade R-CNN	−3.17 [†]
EuroCity-Night	YOLOX	−4.35 [†]
EuroCity-Night	RetinaNet	−4.34 [†]
EuroCity-Night	Cascade R-CNN	−4.44 [†]
Mean		−3.27

**Fig. 2.** Phase 1 LP audit: gender gaps (a) and age gaps (b) across four datasets and three detectors. Asterisks mark $p < 0.05$.

4.2 DWBBR Intervention

On the training set, DWBBR behaves as intended (Figure 3): the female-to-male CIoU gap narrows from -2.82 pp at epoch 1 to -1.50 pp at epoch 50 (46.8% relative reduction), while the female loss weight drifts toward parity as the per-group IoU ratio improves.

On held-out data the picture changes (Table 4, Figure 4). Under the primary seed, DWBBR female CIoU is 0.7715 against baseline 0.7790; the female-to-male gap widens from -1.52 pp (baseline) to -2.74 pp (DWBBR). A one-sided Mann–Whitney test finds no significant female improvement ($p = 0.71$). Across three seeds the mean validation gap moves from -1.78 ± 0.48 pp (baseline) to -1.36 ± 0.63 pp (DWBBR), a 23.6% mean reduction accompanied by higher variance and unstable best-checkpoint epochs ($\{15, 17, 21\}$ for DWBBR versus

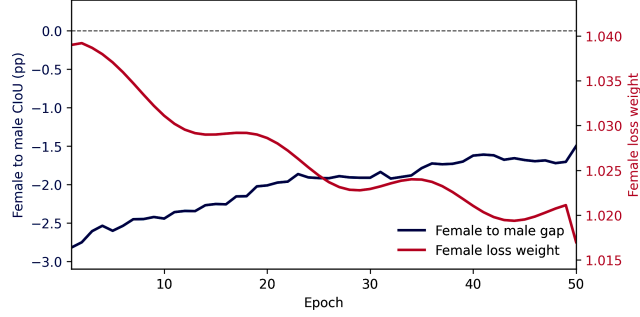


Fig. 3. DWBBR training dynamics: LP gap convergence over 50 epochs on BDD100K.

{34, 17, 24} for baseline). DWBBR therefore closes the training LP gap, yet the improvement does not transfer cleanly: variance rises, checkpoint selection becomes unstable, and under the primary seed the validation gap worsens. This aligns with known limits of group reweighting under train–test demographic shift [11]. We report it as a fairness generalization gap rather than as a claim that all mitigations must fail. Figure 5 shows qualitative Localization Parity violations on BDD100K.

Table 4. Validation CIoU by group on the BDD100K validation subset (423 images). Gap = weaker group minus reference group (pp). Multi-seed means use seeds {42, 123, 7}.

Group	Model	n	CIoU	95% CI	Gap (pp)
Male	Baseline	555	0.7942	–	–
Female	Baseline	376	0.7790	[0.768, 0.790]	–1.52
Male	DWBBR	555	0.7990	–	–
Female	DWBBR	374	0.7715	[0.760, 0.783]	–2.74
Mean female to male, baseline					-1.78 ± 0.48
Mean female to male, DWBBR					-1.36 ± 0.63

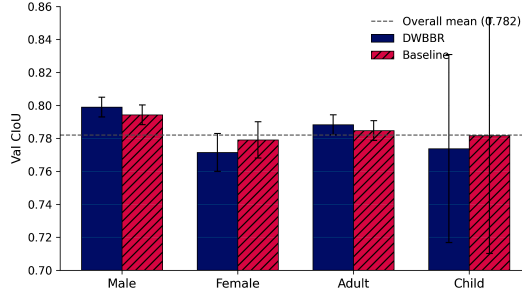


Fig. 4. Validation-set CIoU for DWBBR and the baseline, by demographic group.



Fig. 5. Qualitative Localization Parity violations on BDD100K. Female pedestrians more often receive looser predicted boxes than male pedestrians in comparable scenes.

Demographic labels in BDD100K are third-party visual inferences with binary gender, no self-identification, and undocumented annotation guidelines, raising consent and inclusivity concerns [12]. LP is a proxy for box-quality fairness rather than a guarantee of equal downstream risk, and fairer localization can reduce collision risk while also improving re-identification. We publish because we judge the near-term safety benefit to outweigh that dual-use risk. The study examines only gender and age; race, skin tone, disability, body type, and intersectional analyses remain open.

5 Conclusion

Localization Parity addresses a previously unmeasured fairness gap in pedestrian detection: after a person is found, existing audits ignore how tightly the box fits, even though box geometry drives distance, trajectory, and braking estimates. This paper contributes a formal LP criterion, the first confounder-controlled

multi-dataset and multi-detector audit of continuous IoU disparities on gender and age, and a demographic-weighted regression intervention. Across four datasets and three detector families, female and child pedestrians receive systematically lower IoU boxes (mean gaps -1.40 pp and -3.27 pp), and the gaps survive scale stratification, confirming that the disparity is not explained by object size alone.

Future work should evaluate DWBBR across datasets and architectures, adopt fairness-aware checkpoint selection, compare stronger DRO and oversampling baselines, extend audits to skin tone and intersectional groups with higher-quality labels, and quantify how 1–3 pp IoU gaps propagate into planning decisions.

Acknowledgments. We thank the PanAfriCon AI 2026 Program Committee and reviewers for constructive feedback.

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